

VENDOR DATA SYNCHRONIZATION

ONGC SAP S/4HANA → OVL SAP S/4HANA

Interface Design & Data Transfer Document

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1. Objective

The objective of this document is to describe the design, selection logic, payload structure, and process flow for the automated synchronization of Employee Vendor Master data (Account Group : VEMP) from the ONGC SAP S/4HANA system to the OVL SAP S/4HANA system via PI/PO and SAP Cloud Platform Integration (CPI).

Key goals:

- Ensure OVL always holds up-to-date vendor master records that are maintained in ONGC for Employee Vendors.
- Synchronize only changed/delta records (not a full load) to minimize data volume and processing time.
- Maintain data integrity across both systems with a reliable, auditable interface.



2. Scope

2.1 In-Scope

- Vendor master records belonging to Account Group VEMP in the ONGC S/4HANA system.
- General Data (LFA1) – Address, Name, Communication.
- Company Code Data (LFB1) – Payment Terms, Reconciliation Account etc.
- Bank Details (LFBK) – Bank Country, Bank Key, Bank Account Number.
- Delta records only – Employee Vendors whose Business Partner record was created / changed on the previous calendar day.
- Only Employee Vendors that exist in OVL (LFB1-BUKRS = OVL) will be updated.
- Initial / Full load migration of Employee Vendor Master data for OVL Company Code.

2.2 Out-of-Scope

- Purchasing Organization data (LFM1, LFM2).
- Withholding Tax data.
- Account Groups other than VEMP.



3. System Landscape

Attribute	Source System	Integration Middleware	Target System
System Name	ONGC SAP S/4HANA	ONGC SAP PI/PO + OVL SAP CPI	OVL SAP S/4HANA
System Type	SAP S/4HANA On-Premise	SAP BTP Integration Suite	SAP S/4HANA PCE
Role	Sender / Source	Orchestration & Transformation	Receiver / Target
Data Extraction	ABAP Batch Job / API	inflow	SOAP / REST API / iDoc



4. Abbreviations & Glossary

Term / Abbreviation	Description
ONGC	Oil and Natural Gas Corporation
OVL	ONGC Videsh Limited
CPI	SAP Cloud Platform Integration (SAP Integration Suite)
BP	Business Partner
VEMP	Vendor Account Group – the specific group under consideration
LFA1	Vendor General Data table
LFB1	Vendor Company Code Data table
LFBK	Vendor Bank Details table
SY-DATUM	SAP system date variable (current date)
PCE	Private Cloud Edition



5. Interface Overview

Interface Name	ONGC-OVL Vendor Master Sync
Interface ID	OVL-INT-VEN-001
Direction	ONGC → OVL (Unidirectional)
Middleware	SAP CPI / SAP Integration Suite
Message Format	JSON / XML (to be confirmed during technical design)
Communication Protocol	REST / SOAP API
Trigger	Scheduled Batch Job (ONGC side)
Frequency	Daily – configurable (recommended: night batch, post midnight)
Error Notification	Email alert / CPI monitoring dashboard

6. Synchronization Schedule

The synchronization will be triggered by a scheduled ABAP Batch Job on the ONGC S/4HANA system.

Parameter	Value
Job Frequency	Daily
Recommended Run Time	After midnight (e.g. 01:00 AM)
Delta Date Logic	BUT000-CHDAT = SY-DATUM - 1 (previous calendar day)
Job Owner	ONGC Basis / IT Operations Team



7. Data Selection Logic

The following steps describe the sequential data selection logic executed on the ONGC S/4HANA system to identify the delta vendor records to be synchronized.

Step 1 – Identify Changed Business Partners

Table: BUT000 (Business Partner Master)

Select all BUT000-PARTNER where:

Condition	Filter Logic	Remarks
Preferred (Delta – Prior Day)	BUT000-CHDAT = SY-DATUM - 1	Batch Job triggered on next calendar day after change
Alternative (Same-Day)	BUT000-CHDAT = SY-DATUM	Batch Job triggered on same calendar day as change

Step 2 – Filter by Vendor Account Group VEMP

CDS View: I_BUSINESSPARTNERSUPPLIER

Pass the BUT000-PARTNER values retrieved in Step 1 to I_BUSINESSPARTNERSUPPLIER-BUSINESSPARTNER. Retrieve I_BUSINESSPARTNERSUPPLIER-SUPPLIER where I_BUSINESSPARTNERSUPPLIER-SUPPLIERACCOUNTGROUP = 'VEMP'.

Step 3 – Validate Vendor Exists in OVL Company Code

Table: LFB1 (Vendor Company Code Data)

Pass I_BUSINESSPARTNERSUPPLIER-SUPPLIER → LFB1-LIFNR and filter where LFB1-BUKRS = OVL . Only vendors with an existing LFB1 entry for the OVL company code will proceed to the next step. New vendor creation is out-of-scope.

Step 4 – Retrieve Change Document Header

Table: CDHDR (Change Document Header)

For each validated Supplier (LFB1-LIFNR), pass the value to CDHDR-OBJECTID and retrieve CDHDR-CHANGENR where CDHDR-OBJECTCLAS = 'KRED'. This gives the list of all change documents raised for these vendors.







Step 6 – Prepare Payload

Using the LIFNR values from Step 3, read the latest data from LFA1, LFB1, and LFBK and prepare the outbound payload as described in Section 8.



8. Payload Structure

The outbound payload sent from ONGC to OVL via CPI will contain the following fields grouped by SAP table / segment.

8.1 LFA1 – Vendor General Data

Field Name	Description	Key?	Remarks
LIFNR	Vendor Account Number	**Yes**	Primary Key – 10-char alphanumeric
LAND1	Country Key	No	ISO 2-char country code
NAME1	Name 1 of Vendor	No	
ORT01	City	No	
ORT02	District / P.O. Box City	No	
PSTLZ	Postal Code	No	
REGIO	Region / State Code	No	
SORTL	Sort Field	No	Short name for search
STRAS	Street / House Number	No	
ADRNR	Address Number	No	Links to ADRC table
MCOD1	Search Term 1	No	Uppercase first name/org
MCOD3	Search Term 3	No	Uppercase address/city
ANRED	Title	No	Salutation code
KONZS	Group Key	No	Corporate group assignment
SPRAS	Language Key	No	ISO language code
TELF1	Telephone 1	No	
GBORT	Place of Birth	No	Natural person only
J_1IPANNO	PAN Number (India)	No	India-specific tax field



8.2 LFB1 – Vendor Company Code Data

Field Name	Description	Key?	Remarks
LIFNR	Vendor Account Number	**Yes**	FK to LFA1-LIFNR
BUKRS	Company Code	**Yes**	OVL Company Code – used as filter in Step 3
PERNR	Personnel Number	No	Employee linked as vendor
ZUAWA	Key for Sorting According to Assignment	No	
AKONT	Reconciliation Account	No	G/L Account
ZWELS	Payment Methods	No	Allowed payment method codes
ZTERM	Payment Terms Key	No	
KVERM	Memo / Clerk's Notes	No	Free text

8.3 LFBK – Vendor Bank Details

Field Name	Description	Key?	Remarks
LIFNR	Vendor Account Number	**Yes**	FK to LFA1-LIFNR
BANKS	Bank Country Key	**Yes**	ISO 2-char country code
BANKL	Bank Key (Routing/BIC)	**Yes**	
BANKN	Bank Account Number	No	Encrypted in S/4HANA
BVTYP	Partner Bank Type	No	Identifier for multiple banks
KOINH	Account Holder Name	No	If different from vendor name



9. Process Flow

The end-to-end flow is described below:

Step	Activity	Description
1	ONGC Batch Job Triggered	Scheduled ABAP Batch Job fires on ONGC S/4HANA at pre-configured time (e.g. 01:00 AM).
2	Delta Record Identification	Job reads BUT000 for records where CHDAT = SY-DATUM - 1 (previous day).
3	Account Group Filter	BP numbers are passed to I_BUSINESSPARTNERSUPPLIER; only SUPPLIERACCOUNTGROUP = VEMP records are retained.
4	OVL Existence Check	Filtered Suppliers are checked against LFB1 for the OVL Company Code (BUKRS). Non-existent records are discarded (out-of-scope).
5	Payload Preparation	ABAP program reads current values from LFA1, LFB1 (OVL BUKRS), LFBK and builds outbound message.
6	CPI iFlow – Message Routing	ONGC sends the payload to CPI iFlow via REST/SOAP. CPI performs message validation and transformation.
7	Target API Call – OVL S/4HANA	CPI calls OVL S/4HANA API / BAPI_VENDOR_CHANGE (or equivalent) to update the Business Partner for Employee Vendor record.
8	Acknowledgement & Logging	OVL returns success/failure acknowledgement. CPI logs the result. Errors trigger alerts.



10. Error Handling & Exception Management

Error Scenario	Detection Point	Handling Approach	Responsible Team
No delta records found	ONGC Batch Job	Job exits gracefully; logs 'No records to sync' message.	
Vendor not found in OVL	Step 3 – LFB1 check	Record discarded; logged as 'Skipped – Vendor not in OVL'.	
CPI connection failure	ONGC → CPI call	Retry 3 times with exponential backoff; alert sent if all retries fail.	
Payload validation failure	CPI iFlow	Message moved to error queue; alert sent to interface monitor.	
OVL API rejection	CPI → OVL call	Error response logged; record flagged; manual resolution required.	







11. Security & Authorization

- Dedicated RFC / API user IDs must be created in both ONGC and OVL systems with minimal required authorizations.
- Initial / Full load migration of Vendor Master data.
- Access to CPI iFlow design artifacts must be restricted to the interface development team.



12. Testing & Sign-Off

12.1 Testing Phases

Phase	Description	Responsible	Status
Unit Testing	Individual ABAP program / CPI iFlow component testing	Development Team	
Integration Testing	End-to-end flow from ONGC Batch Job → CPI → OVL update	Development + Functional	
Performance Testing	Verify batch job handles expected record volumes	Development + Basis	
UAT	Business validation of synchronized vendor data in OVL	Business / Key Users	
Regression Testing	Ensure no impact to existing ONGC / OVL processes	Functional Team	

12.2 Sign-Off

Role	Name	Signature	Date
Business Owner – ONGC			
Business Owner – OVL			
IT Lead – ONGC			
IT Lead – OVL			